

Retinal AI Handbook

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July 20, 2026

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Preface

The retina is the only part of the central nervous system that can be directly visualized non-invasively. Owing to its shared embryological origin, anatomical structure, and physiological function with the brain, retinal microvascular alterations have been associated with various systemic diseases. Advances in retinal imaging and artificial intelligence have transformed the retina from an organ primarily examined for ocular diseases into a valuable source of quantitative biomarkers for systemic health. Recent advances in artificial intelligence (AI) have further enhanced the potential of retinal imaging as a non-invasive biomarker for cerebrovascular health. These developments have transformed retinal image analysis from a purely qualitative clinical assessment to large-scale, quantitative, and automated evaluation of retinal images, enabling early detection of disease, risk stratification, and monitoring of disease progression.

The motivation for this handbook originated from my own experience as a Computer Science PhD student entering the field of retinal imaging. Although building machine learning models was familiar, understanding biomedical terminology, retinal anatomy, clinical concepts, and disease mechanisms proved to be a much greater challenge than expected. Many important concepts were scattered across ophthalmology textbooks, neuroscience references, epidemiology papers, and artificial intelligence literature. This fragmentation made it difficult to build a coherent understanding of the field. This handbook was therefore developed as a living knowledge base that helps translate complex clinical concepts into computational understanding. It is intended to evolve alongside scientific advances, serving as a living knowledge base for researchers, clinicians, and students exploring the intersection of retinal imaging, vascular biology, and artificial intelligence. The handbook is organized into chapters, each focusing on a specific topic, with a glossary of terms provided for reference. The content is written in English, with Vietnamese translations provided for selected glossary terms to facilitate understanding for Vietnamese-speaking readers. Rather than providing definitive answers, this handbook aims to support continuous learning, critical thinking, and interdisciplinary research throughout the rapidly evolving field of retinal imaging and artificial intelligence.

How to read this handbook

Purpose of This Handbook

The primary purpose of this handbook is to facilitate interdisciplinary learning by organizing retinal imaging knowledge into a structured knowledge base. Rather than providing definitive answers or isolated facts, this handbook emphasizes the relationships among biological concepts, clinical reasoning, biomedical terminology, and computational methods. By progressively connecting these concepts, the handbook aims to help readers develop a coherent understanding of retinal imaging research, allowing them to interpret scientific literature, communicate across disciplines, and conduct evidence-informed research.

Scope

This handbook focuses on the foundational concepts required to understand retinal imaging research from both biomedical and computational perspectives. This edition focuses primarily on color fundus photography (CFP), retinal microvasculature, retinal biomarkers, and their applications in systemic diseases, particularly stroke and atherosclerosis. To maintain conceptual consistency, this handbook prioritizes depth over breadth, focusing on concepts that form the foundation of retinal imaging research.

Future editions will progressively expand the handbook to include additional retinal imaging modalities, including optical coherence tomography (OCT), OCT angiography (OCTA), and fluorescein angiography (FA), together with multimodal retinal imaging, foundation models, and emerging computational approaches for retinal image analysis. Future editions will also explore the relationships between retinal imaging and systemic diseases, including diabetes, hypertension, and neurodegenerative disorders.

As retinal imaging research continues to evolve, the scope of this handbook will be updated accordingly to incorporate emerging scientific evidence and technological advances.

Organization of This Handbook

The handbook is organized around concepts rather than individual research papers or algorithms. Each chapter introduces a coherent group of related concepts, allowing readers to progressively build an integrated understanding of retinal imaging research. Follow that, each concept is presented using a consistent structure, including a definition, biological background, clinical relevance, computational relevance, related terminology, and suggested readings. This standardized organization allows readers to easily navigate different topics efficiently while maintaining conceptual continuity throughout the handbook.

Glossary terms and cross-references are provided throughout the handbook to encourage connections between related concepts rather than isolated memorization. Selected references are also included to guide readers towards the original scientific evidence and support further exploration of each topic.

How to Use This Handbook

To obtain the greatest benefit from this handbook, readers are encouraged to approach the material with an interdisciplinary mindset, integrating perspectives from biology, medicine, and computational sciences. Although the chapters are organized sequentially to facilitate progressive learning, individual chapters and concepts may also be consulted independently through the glossary and cross-references.

Table 1: Recommended Reading Pathways

Reader	Suggested Approaches
AI researchers	Focus on biological background, clinical relevance, and retinal biomarkers before studying computational methods.
Biomedical researchers	Begin with retinal anatomy and microvasculature before exploring AI-related chapters.
Clinicians	Focus on imaging modalities, biomarkers, and clinical interpretation.
Students	Read sequentially to establish interdisciplinary foundations.

Readers are encouraged to revisit previously covered concepts as their understanding develops, since many topics in retinal imaging are closely interconnected. As this handbook is intended to be a living knowledge base, constructive feedback, corrections, and suggestions for future editions are always welcome. If you identify any inaccuracies, outdated information, or topics that could be improved, please feel free to contact me at thanhhuong792000@gmail.com. Every contribution, no matter how small, is greatly appreciated.